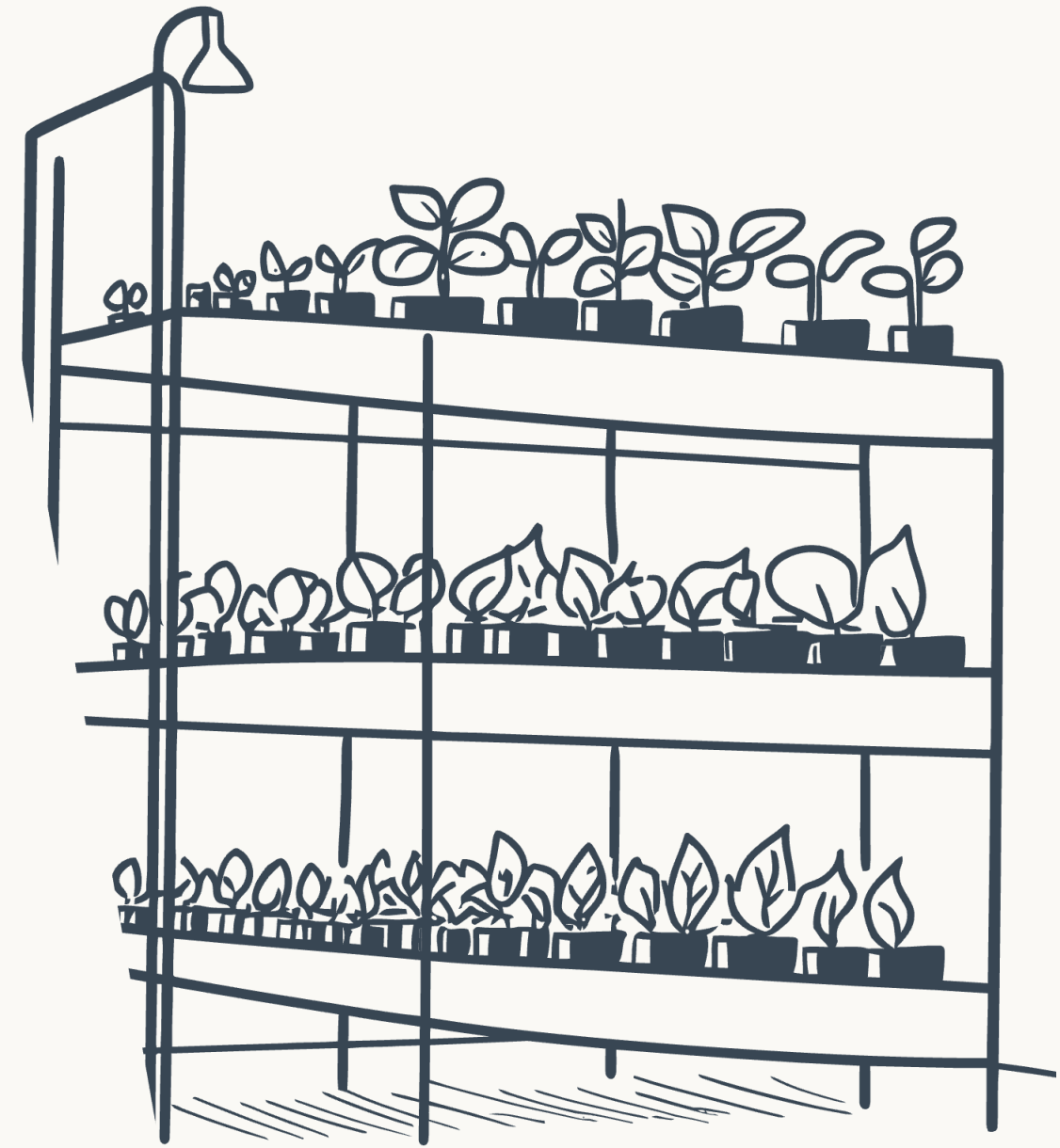




Biological Ground Truth for Autonomous Sensor Validation in Controlled Environment Agriculture

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Modern Agriculture Runs on Sensors

Controlled Environment Agriculture depends on uninterrupted data. Sensors track nutrient chemistry, water quality, and environmental conditions — and the ecosystem lives or dies by their readings

Sensors Degrade

They drift, foul, and fail silently over time

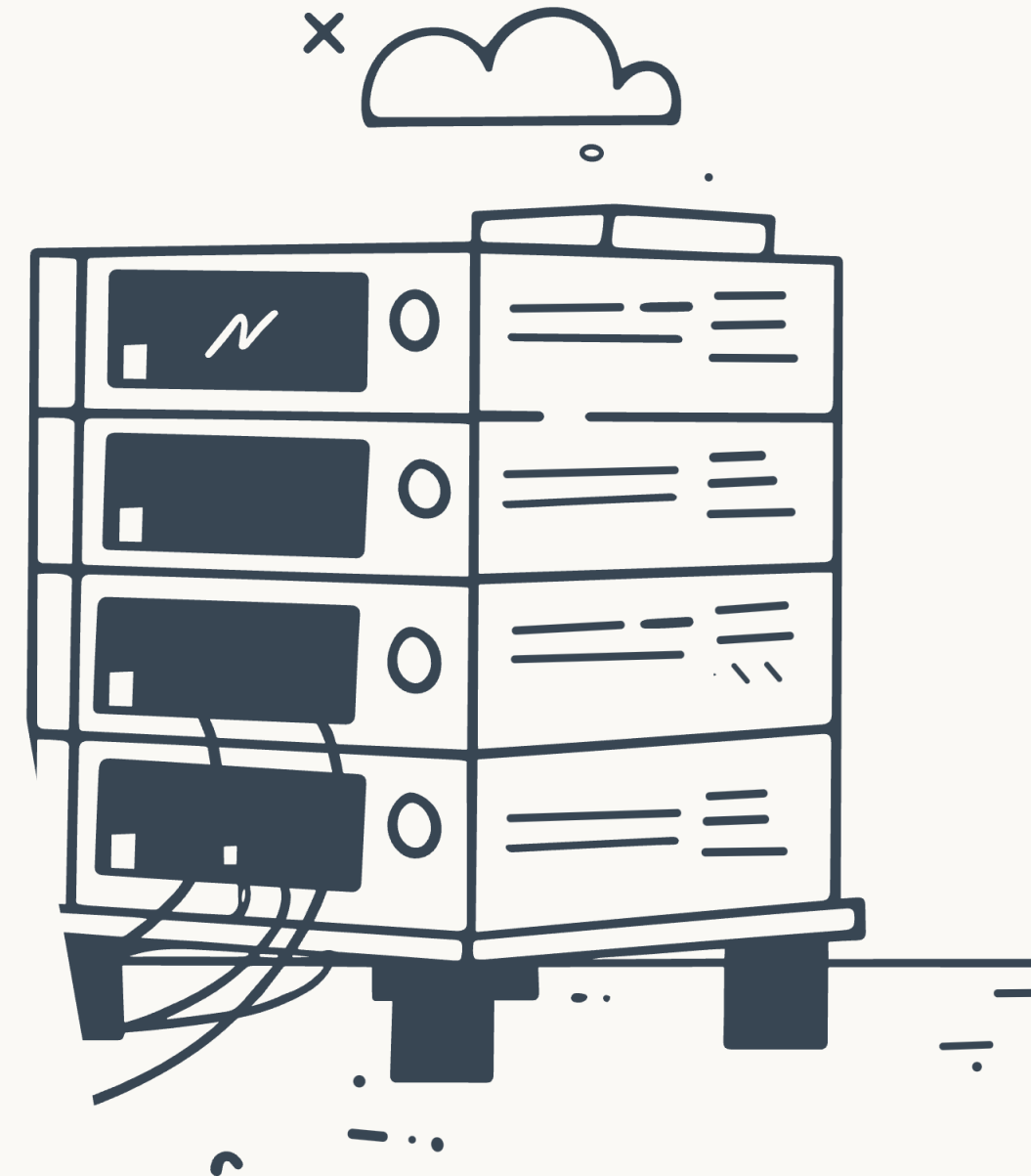
Failures Hide

Damage is detected only after biology collapses

The Fix?

Deploy more hardware, not verify what's already there

We trust sensors.... but we rarely verify them!



A Decade Studying Living Systems

- Built experimental aquaponic systems since early research work
- Bioengineering + Entrepreneurship + Earth Systems background focused on real-world deployment



Working with Biological Data in Practice

Biological data is uniquely challenging:

Noisy

Often contains background interference and irrelevant signals.

Incomplete

Missing values, irregular sampling, and gaps are inherent.

Misleading

Can lead to spurious correlations or misinterpretations if not handled carefully.

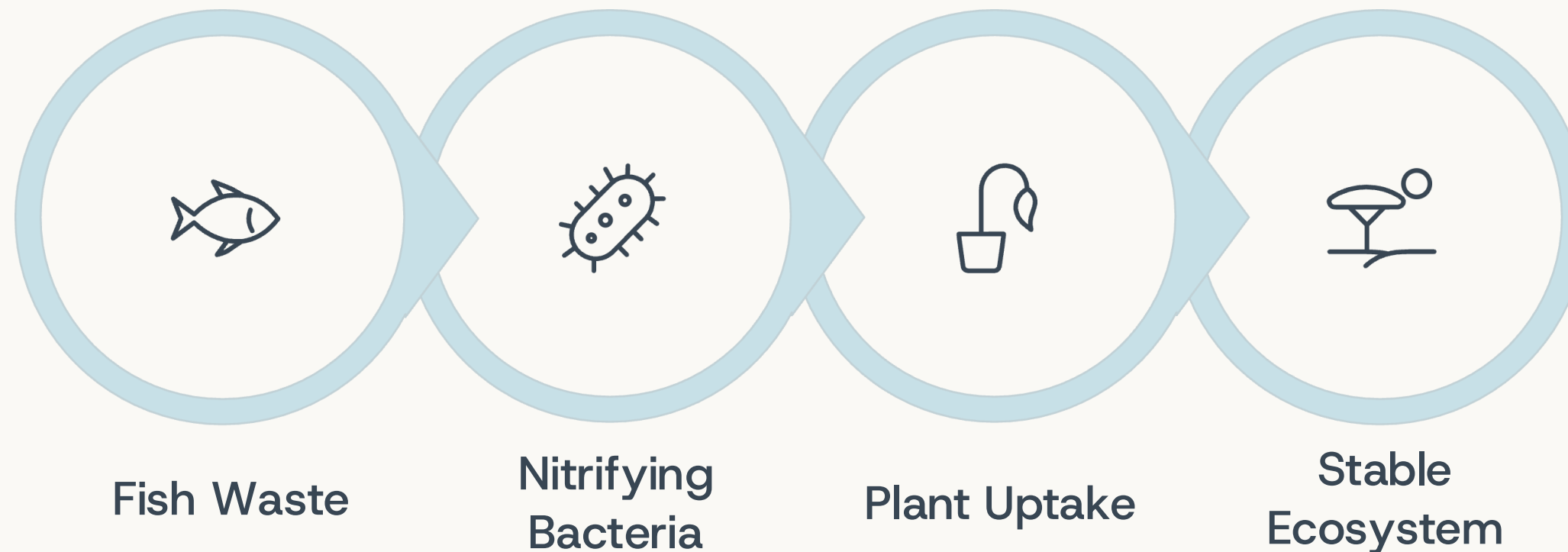
Yet, critical decisions are routinely made based on this imperfect information.

- ① Through my experience, I've observed firsthand that biological data is rarely perfectly reliable, but **decisions are always made based on it...** with significant consequences.

Biology Follows Predictable Patterns

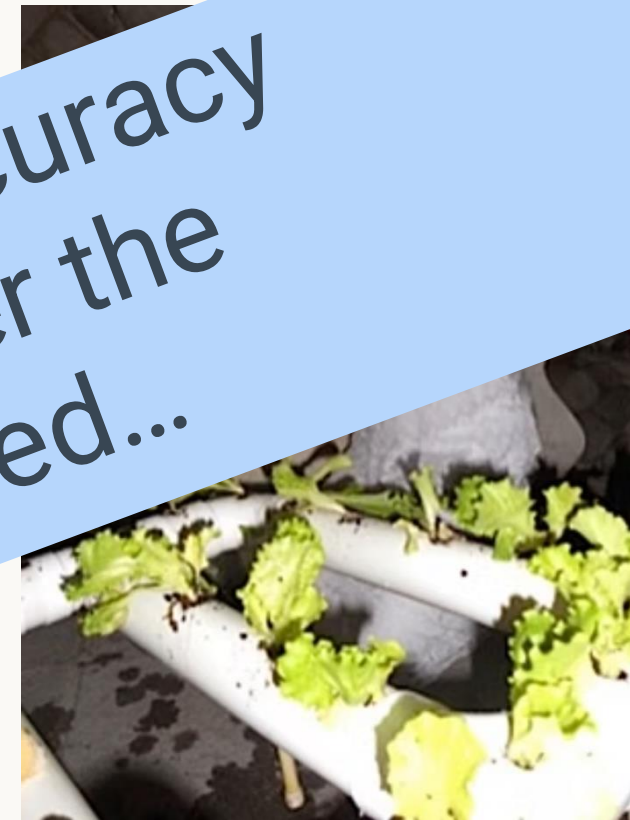
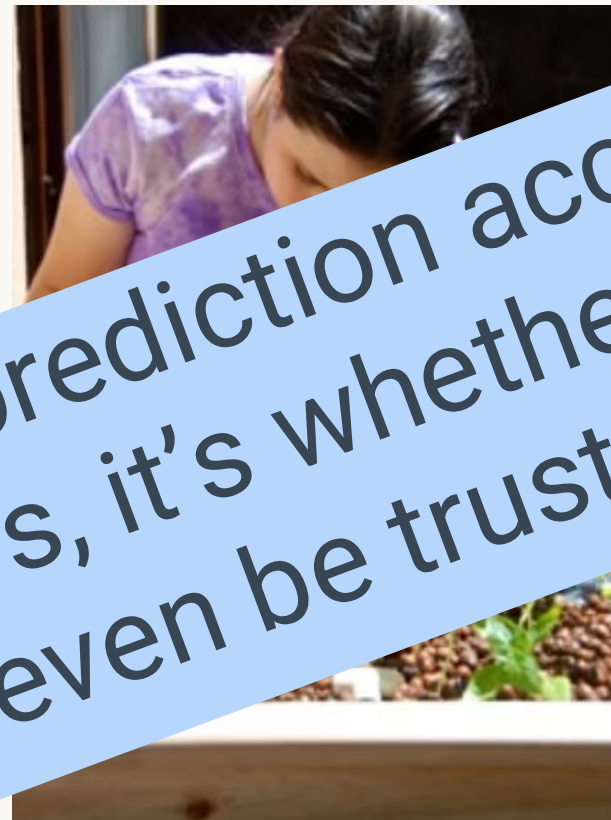
Years of Experimental Observation Reveal a Hidden Structure

Living ecosystems are not random. Healthy aquaponic systems evolve through structured biochemical sequences — and that structure is measurable.



When a sensor reading deviates from the expected biological trajectory, it may not be the biology that's wrong it may be the sensor. **Biology itself can validate sensing systems!**

Real Biological Ground Truth



But the issue is not prediction accuracy with those sensors, it's whether the data itself can even be trusted...

...ments collected manually,
...ectronic sensor, across two full
...aponic systems

Why This Data Is Unique

- Laboratory-grade precision, not electronic estimates
- Captures true ecosystem evolution over time
- Fully independent from the sensors we aim to validate

Data Available: Temperature, pH, Carbonate Hardness (KH), General Hardness (GH), Ammonia, Nitrates, Nitrites

From Problem to System

If sensor data cannot be trusted, how can we validate it?

Old Focus

Improving Sensors

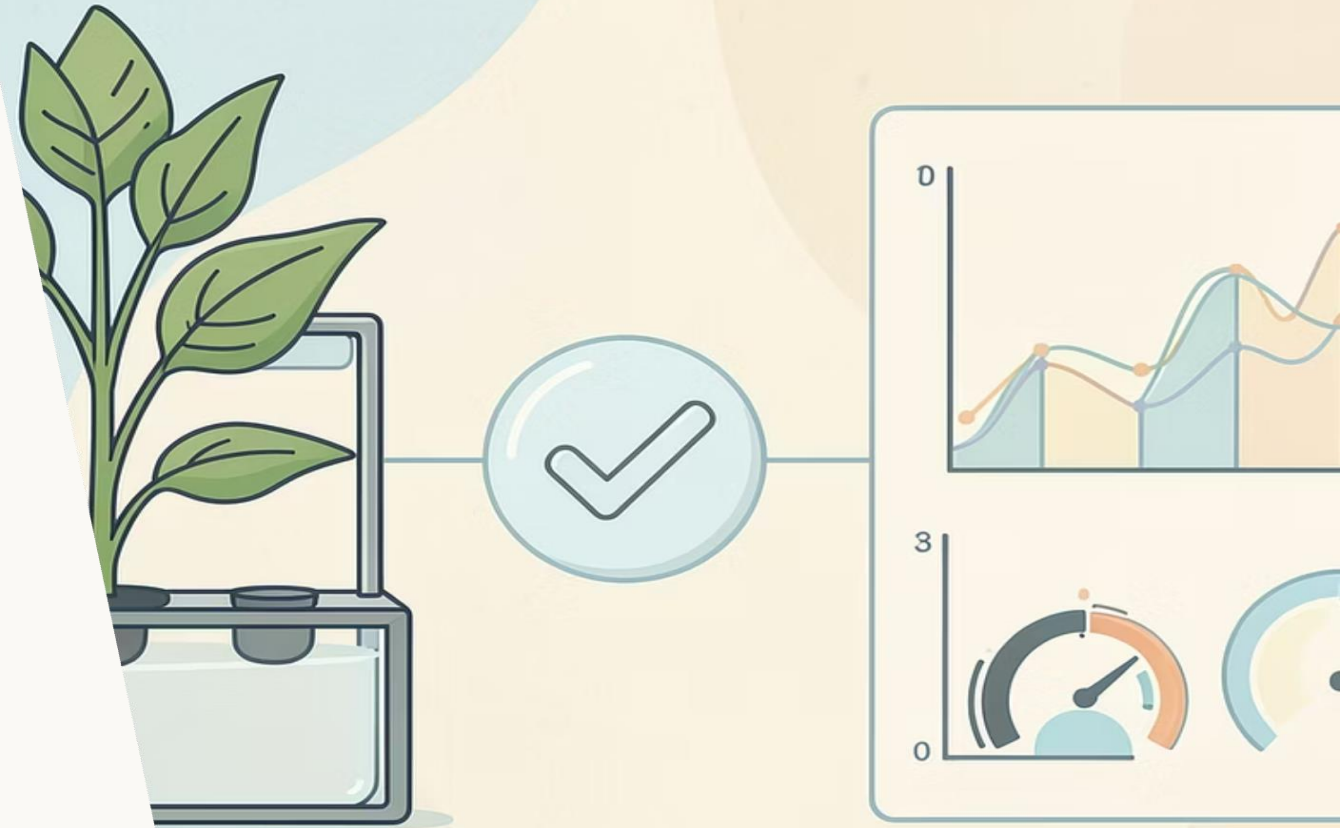
New Focus

Validating Data

The Hydro-Twin Concept

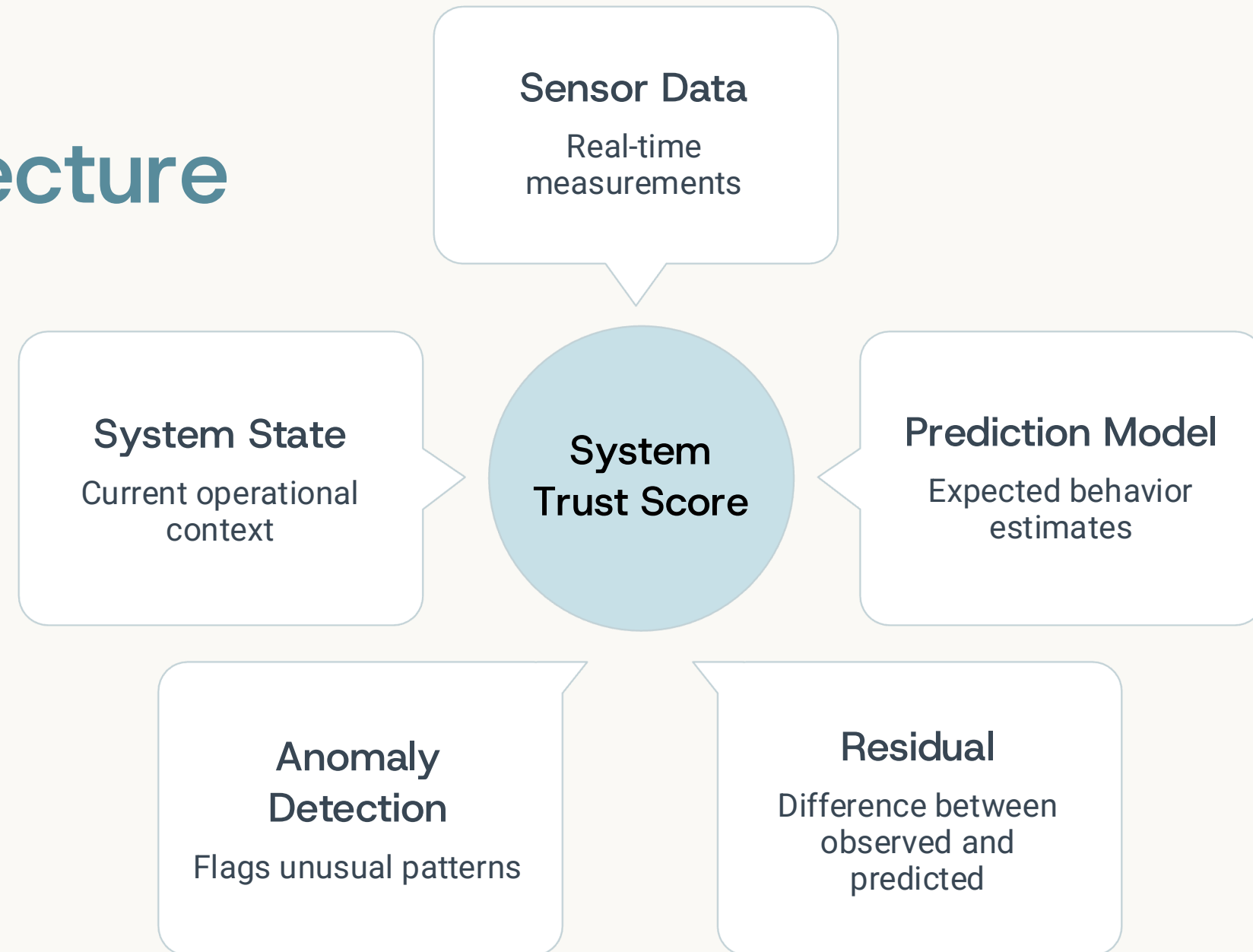
Biological response as
ultimate source of truth

Compare what **we expect from our models** with what the system
actually **observes**, we validate the
process against biological reality



ARCHITECTURE DETAIL

System Architecture



ARCHITECTURE DETAIL

The Prediction Layer

At the core of Hydro-Twin, a sophisticated prediction layer anticipates the expected biological behavior of the system, setting the baseline for validation

Model Type

Utilizes **Random Forest Regression** for robust and accurate predictions, capable of handling complex biological interactions.

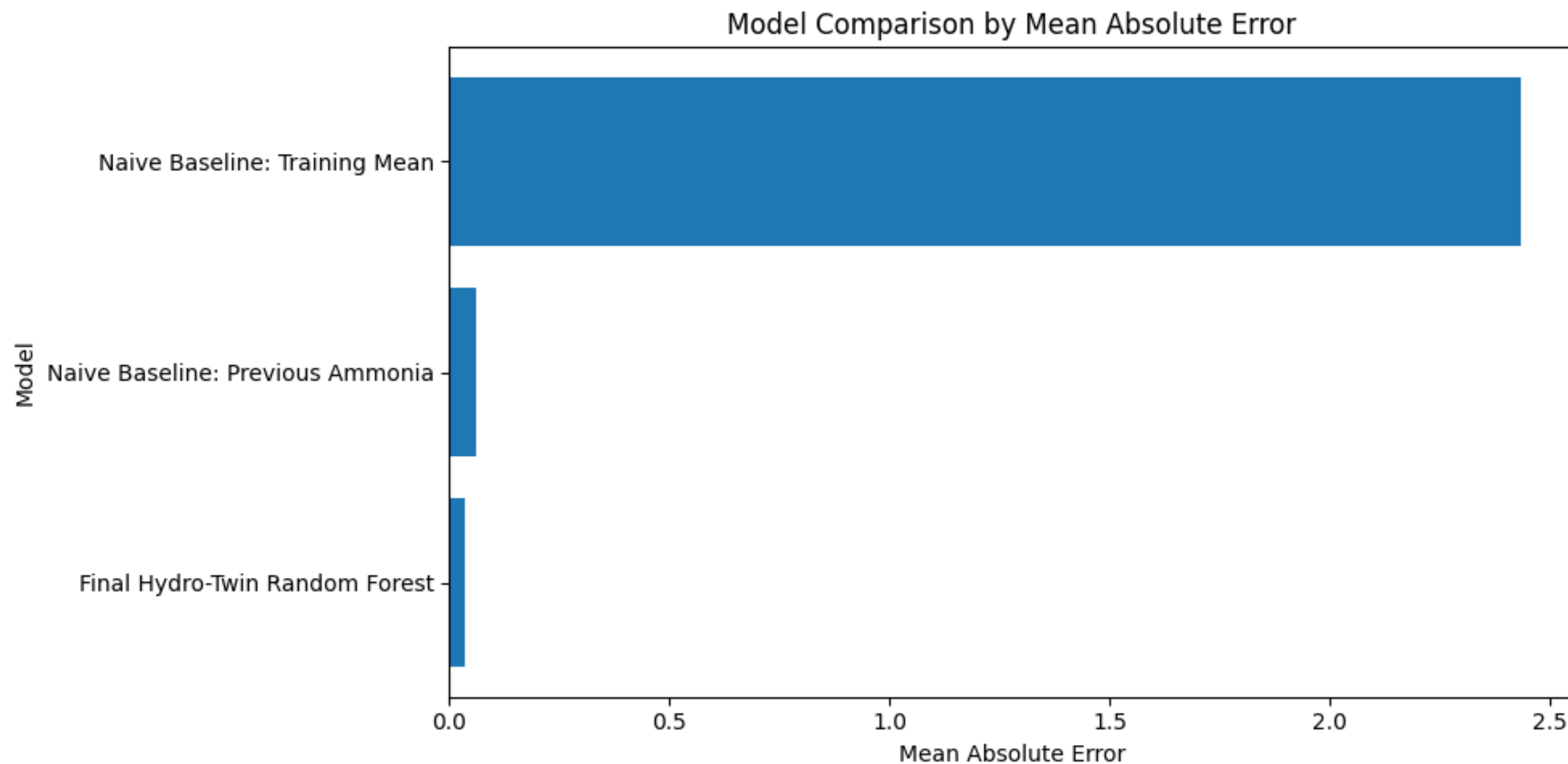
Key Inputs

Leverages critical environmental data including **pH**, **temporal ammonia levels**, and unique **pond identity** for comprehensive analysis.

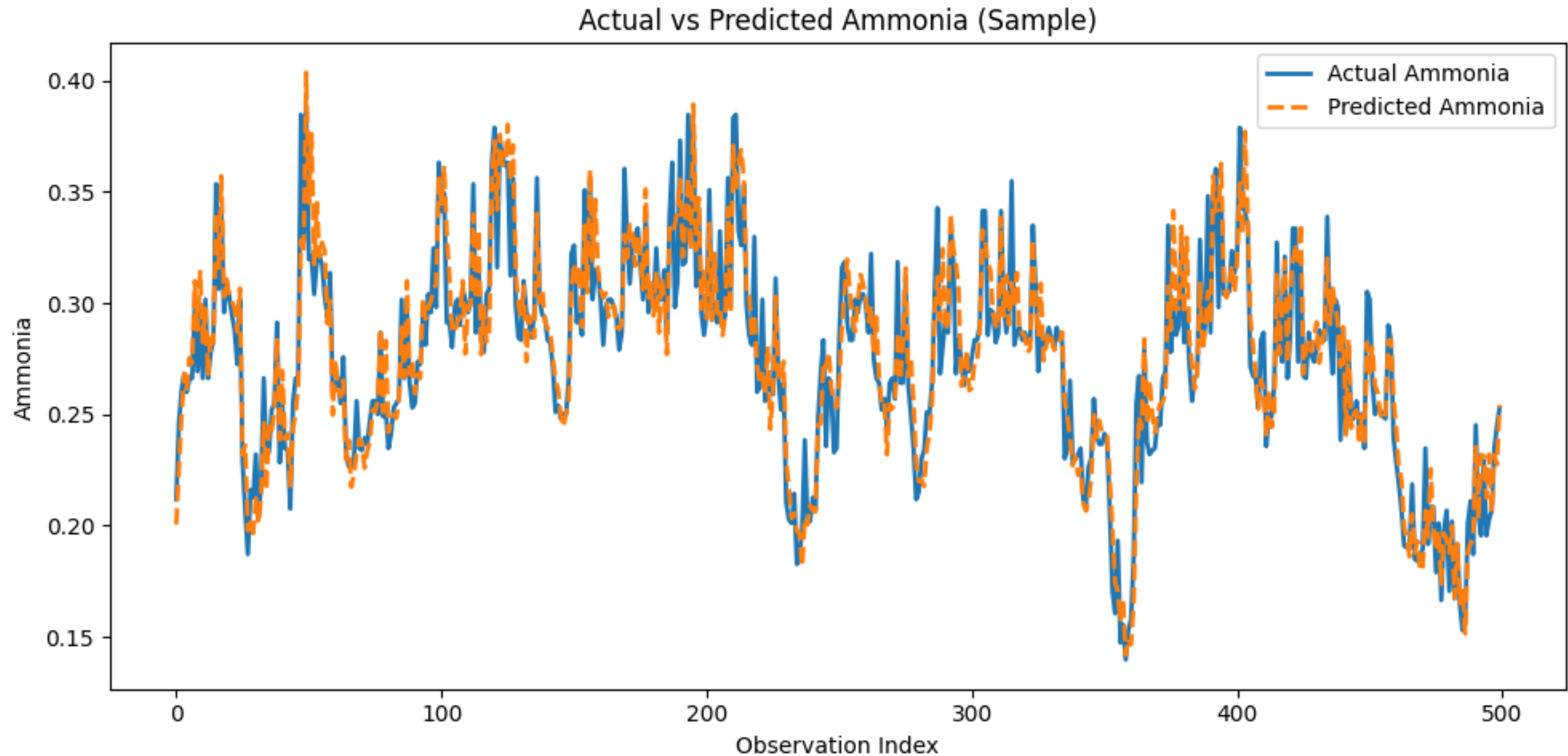
Core Objective

To accurately capture the system's expected biological behavior, establishing a reliable "ground truth" for sensor validation.

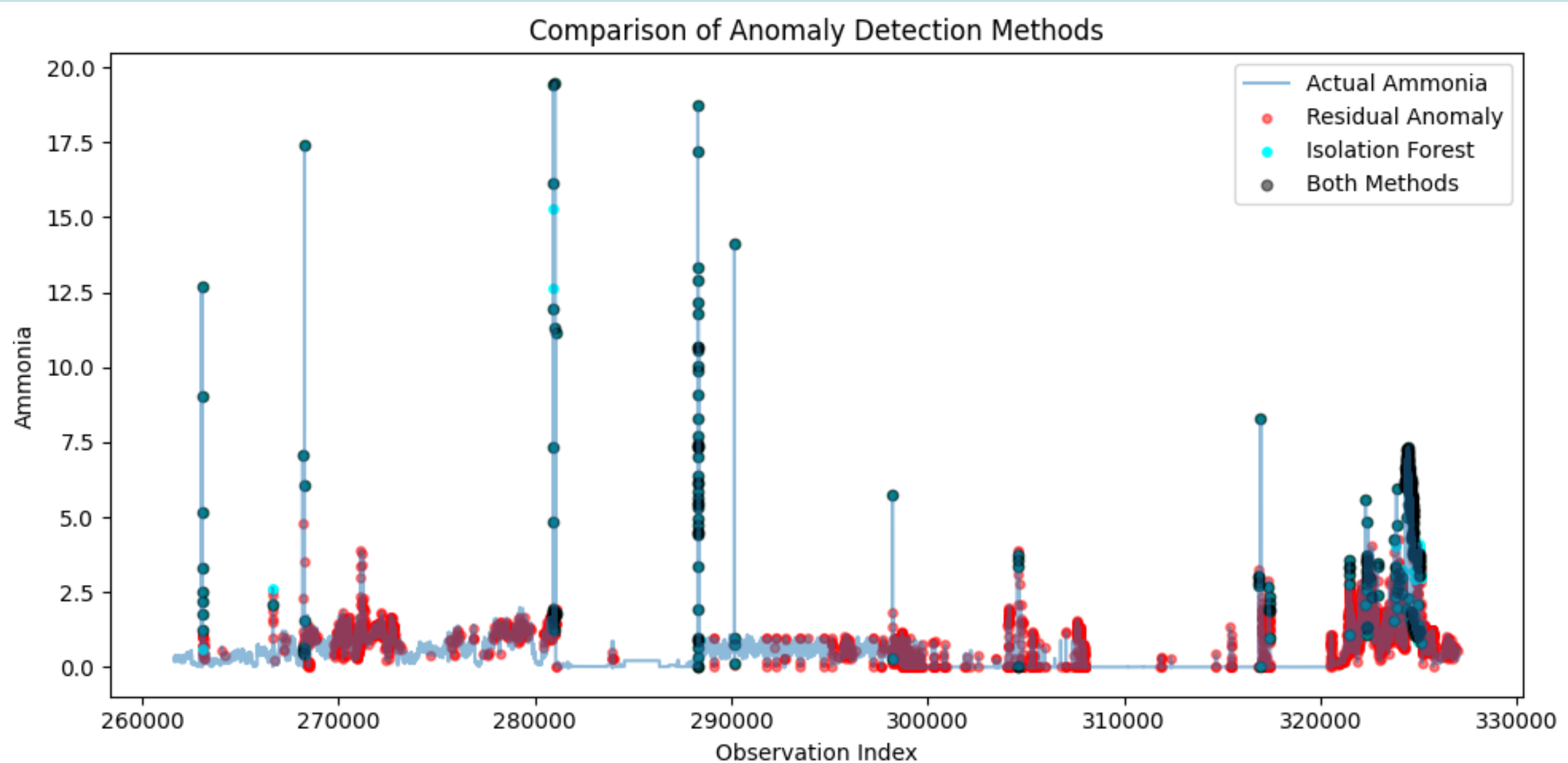
Model Outperforms Baseline



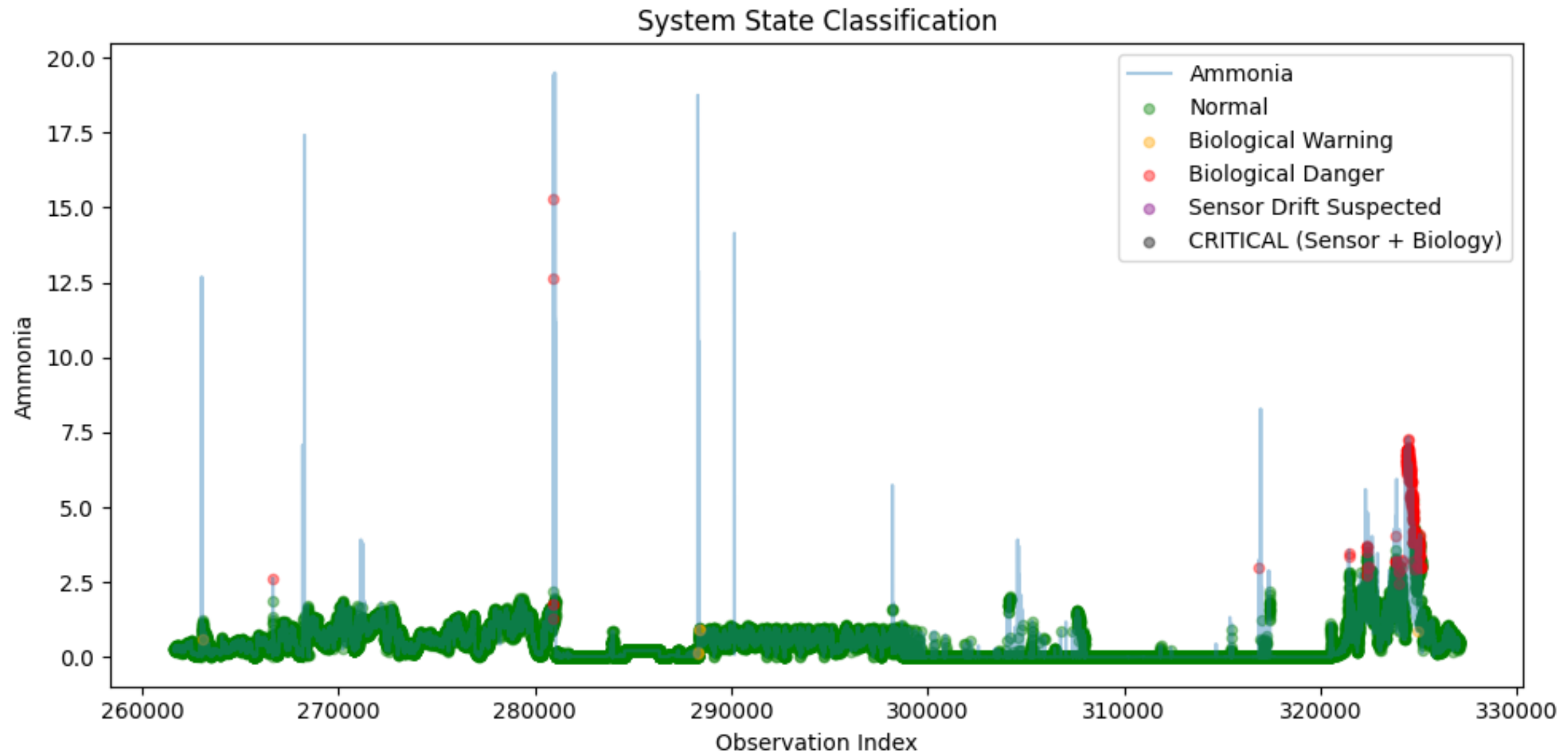
Model Behaviour: Tracking the Biological Signal



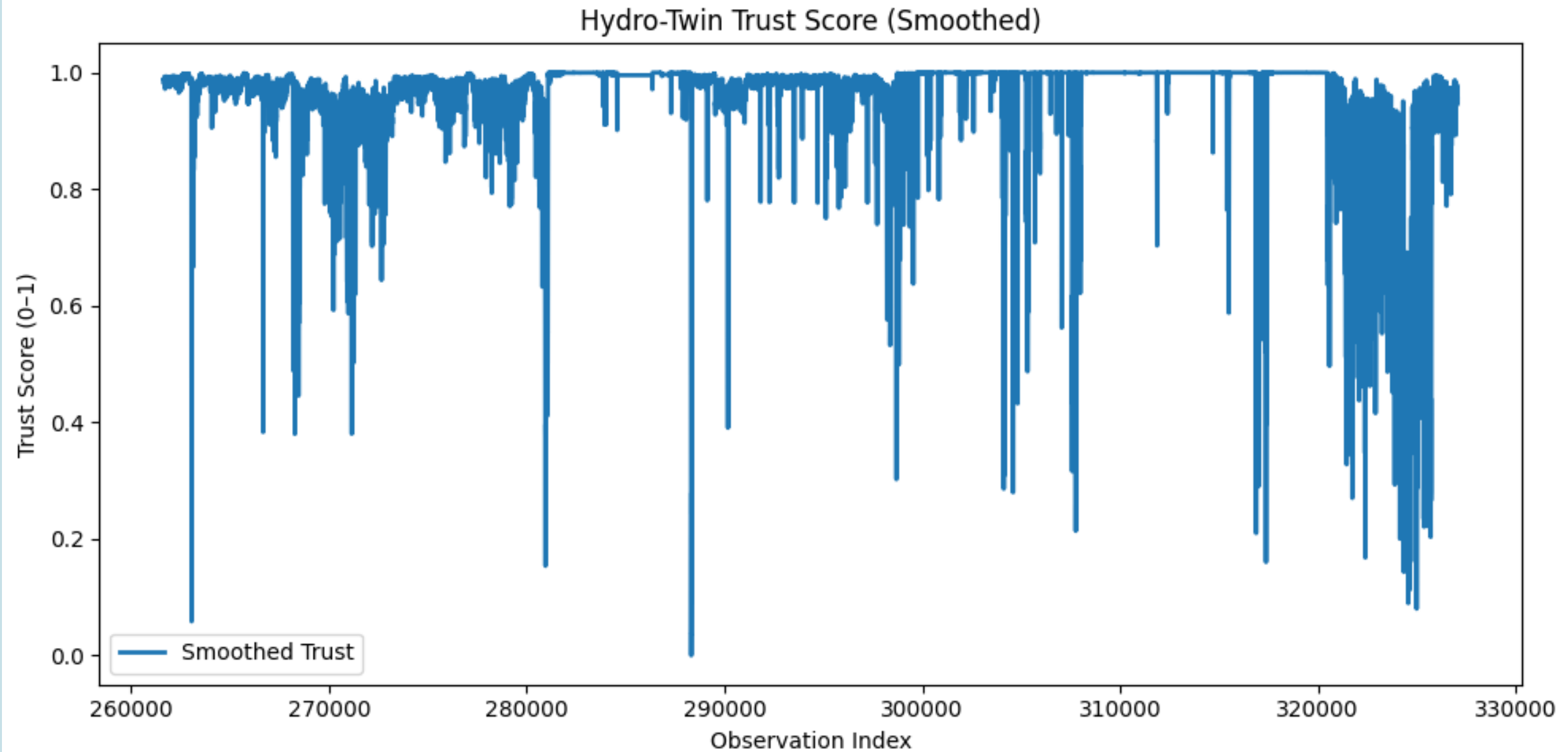
Anomaly Detection: Two Complementary Signals



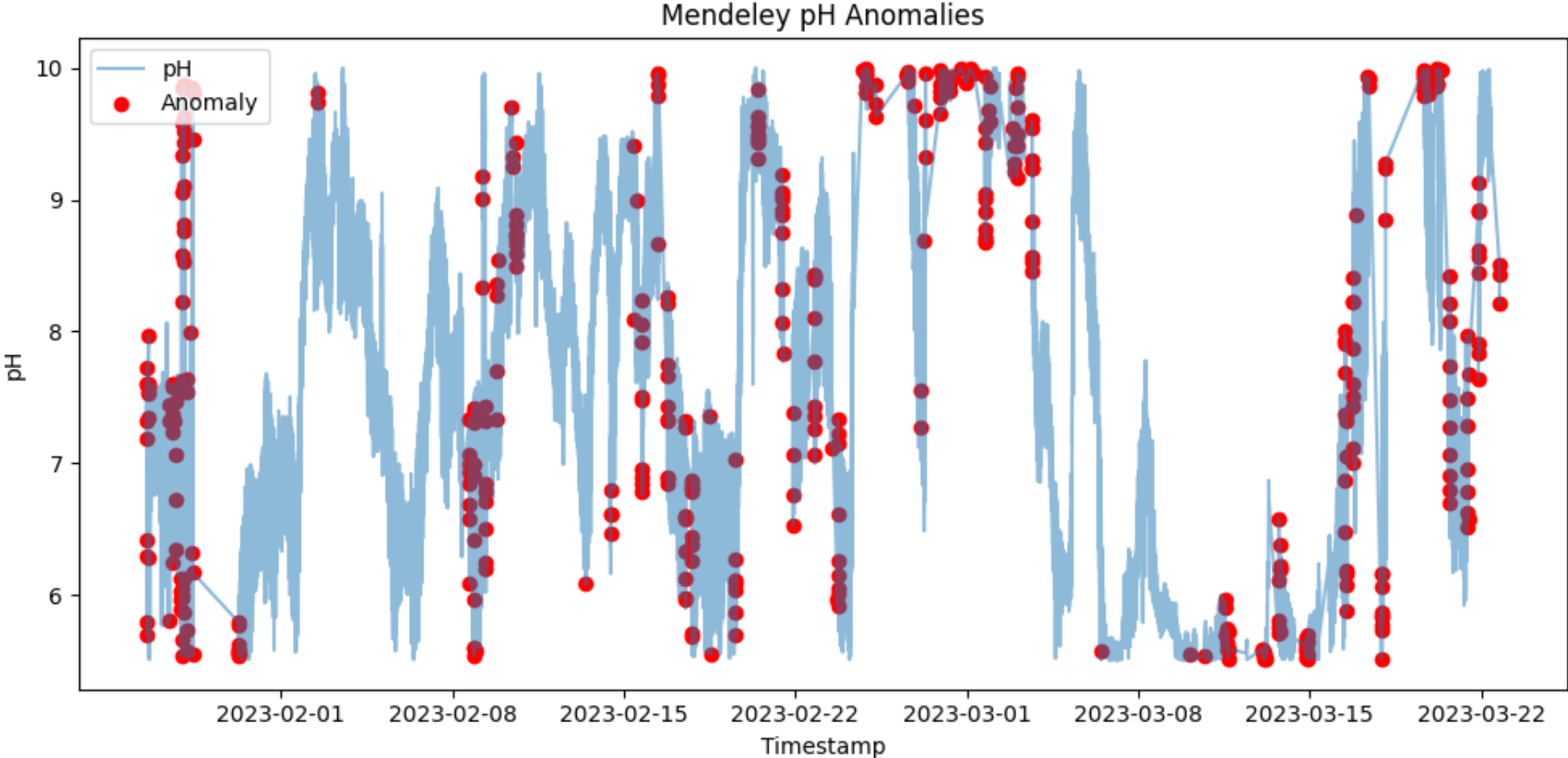
Signal Informed Decision-Making



Can We Trust the Model?



Validation: Mendeley Anomaly



An Interactive System: a Quick Glimpse



System Controls

Select Pond

pond_3

Residual anomaly threshold

0.94

☒ Show residual anomalies

☒ Show Isolation Forest anomalies

☒ Show system states

Residual anomalies flagged: 4138

HydroTwin Monitoring System

Interactive aquaponics monitoring system for prediction, anomaly detection, state classification, and trust scoring.
Developed by Daniela Najmias Lang

Ammonia values are shown in PPM (parts per million)

> How to use this dashboard

Current State	Trust Score	Residual Threshold	Residual Flags
Normal	1.00	0.395	4138

> What do these metrics mean?

Actual vs Predicted Ammonia

Observed vs Expected Ammonia

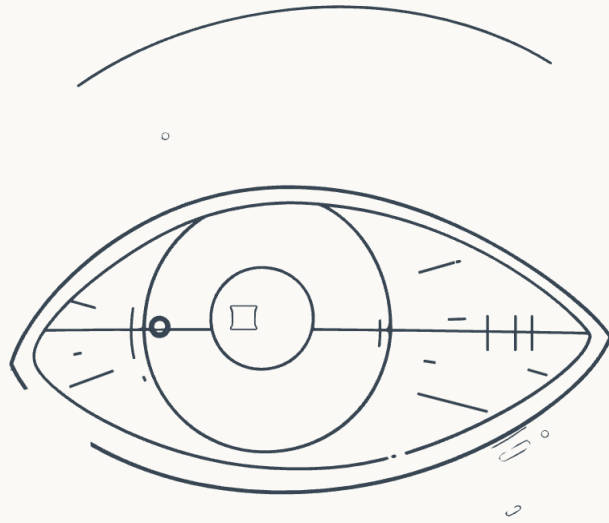
hydrotwin.streamlit.app

The HydroTwin Opportunity

A New Asset Class in Agritech!

Hydro-Twin converts biological data into a validation engine:
**a software layer that sits above existing hardware
AND continuously audits it!**

- No replacement hardware required
- Compatible with existing infrastructure
- Scalable SaaS model potential
- Grounded in real biological data



2

Experimental Reference Studies

Manual lab-grade measurements per system for biological uptake regulation

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Catfish Ponds

IoT aquaponic based system with multisensory and multi-pond information

3

Geo-temporal Systems

Unique world conditions and scenarios showing scalability of the model to multiple controlled environment agriculture scenarios

From Hardware Dependence to Biological Intelligence

Today, farms add sensors to compensate for sensor failure

Hydro-Twin proposes a fundamentally different approach: use biology as the reference system

1

Autonomous Validation

Sensors verified continuously without human intervention

2

Early Failure Detection

Deviations flagged before biological damage even occurs

3

Reduced Hardware Cost

Less reliance on redundant physical monitoring equipment

4

Scalable Intelligence

Applicable across all controlled agriculture environments

Vertical farming, hydroponics, aquaponics, and more!



When biology becomes the reference system, monitoring becomes intelligent